



ROBERT H. SMITH SCHOOL OF BUSINESS

BUDT704: DATA PROCESSING & ANALYSIS IN PYTHON

PREDICTING OBESITY LEVELS USING MACHINE LEARNING

Data Science Project Report

By

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1. INTRODUCTION

Obesity is one of the most critical public health challenges worldwide, contributing significantly to chronic diseases such as cardiovascular disorders, diabetes, hypertension, and metabolic syndromes. Rapid urbanization, sedentary lifestyles, and changes in dietary habits have increased obesity prevalence across age groups and regions. Accurate identification of obesity levels and the factors contributing to them is essential for preventive healthcare and targeted interventions.

The purpose of this project is to analyze demographic, physical, and lifestyle-related data to predict obesity levels using machine learning techniques. Beyond prediction, this study aims to understand the underlying patterns in the data, evaluate the effectiveness of different supervised and unsupervised learning models, and identify the most influential factors contributing to obesity.

The guiding research questions for this project are:

1. Which physical, demographic, and lifestyle factors most strongly influence obesity levels?
2. How effectively do supervised machine learning models predict obesity categories based on physical and lifestyle data?
3. Can unsupervised learning techniques identify natural groupings of individuals based on lifestyle and physical characteristics, independent of predefined obesity labels?

2. DATASET DESCRIPTION

The dataset used in this study is **Estimation of Obesity Levels Based on Eating Habits and Physical Condition**, obtained from the UCI Machine Learning Repository. The data includes individuals from Mexico, Peru, and Colombia.

2.1 Dataset Overview

- Total observations: 2,111
- Total attributes: 17
- Target variable: NObeyesdad
- Data composition:
 - 23% collected from real users via a web platform
 - 77% synthetically generated using SMOTE to balance class distribution

Dataset Reference Link:

<https://archive.ics.uci.edu/dataset/544/estimation+of+obesity+levels+based+on+eating+habits+and+physical+condition>

2.2 Target Variable

The dependent variable, NObeyesdad, consists of seven obesity categories:

- Insufficient Weight
- Normal Weight
- Overweight Level I

- Overweight Level II
- Obesity Type I
- Obesity Type II
- Obesity Type III

The class distribution is relatively balanced, with Obesity Type I being the most frequent class (16.63%) and Insufficient Weight being the least frequent (12.88%). This balance allows fair evaluation across all classes.

2.3 Independent Variables

The independent variables can be grouped as follows:

Numerical Variables (8):

- Age
- Height
- Weight
- FCVC (Frequency of Vegetable Consumption)
- NCP (Number of Meals per Day)
- CH2O (Daily Water Intake)
- FAF (Physical Activity Frequency)
- TUE (Technology Use Time)

Categorical Variables (9):

- Gender
- family_history_with_overweight
- FAVC (Frequent Consumption of High-Calorie Food)
- CAEC (Consumption of Food Between Meals)
- SMOKE
- SCC (Calories Consumption Monitoring)
- CALC (Alcohol Consumption)
- MTRANS (Transportation Method)
- NObeyesdad (Target)

The dataset contains no missing values, simplifying preprocessing.

3. DATA PREPROCESSING

Data preprocessing was conducted to prepare the dataset for analysis and modeling.

3.1 Feature Separation

- Numerical and categorical features were identified using the .info() method.
- Numerical variables were stored in a numerical_columns list.
- Categorical variables were stored in a categorical_columns list.
- The target variable NObeyesdad was separated explicitly.

3.2 Scaling and Encoding

- Numerical features were standardized using StandardScaler to ensure uniform scale across models.
- Categorical variables were encoded using One-Hot Encoding to convert them into numerical format.
- A ColumnTransformer pipeline ensured consistent preprocessing across all models.

3.3 BMI Consideration

BMI was **not included in supervised modeling** to avoid data leakage, as original obesity labels were derived from BMI thresholds. Retaining raw height and weight ensured that the models learned patterns naturally without direct access to label-generating logic.

4. EXPLORATORY DATA ANALYSIS (EDA)

4.1 Target Variable Analysis

The obesity categories are well distributed across the dataset. Higher obesity categories collectively represent a large portion of the population, highlighting the relevance of obesity as a public health issue in the sampled regions.

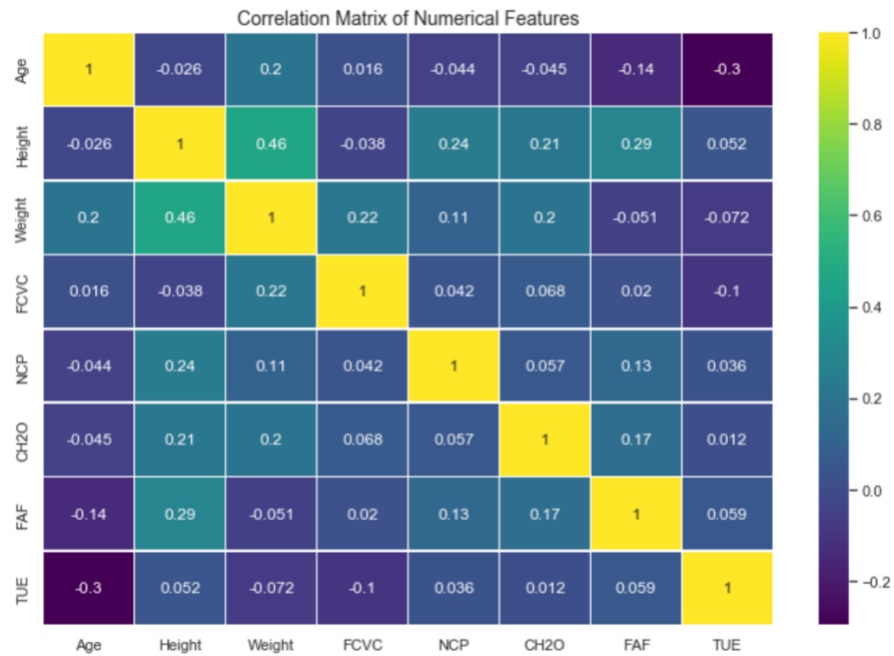
4.2 Numerical Variable Analysis

- **Age:** The dataset is dominated by young adults, with a median age of 22.8 years. Older individuals appear as outliers.
- **Height & Weight:** Height and weight show a strong positive correlation (0.46), as expected.
- **Lifestyle Variables:** FCVC, NCP, CH2O, FAF, and TUE show limited linear correlation with weight individually, suggesting obesity is influenced by multiple interacting factors rather than single behaviors.

4.3 Correlation Heatmap Insights

Key findings from the heatmap include:

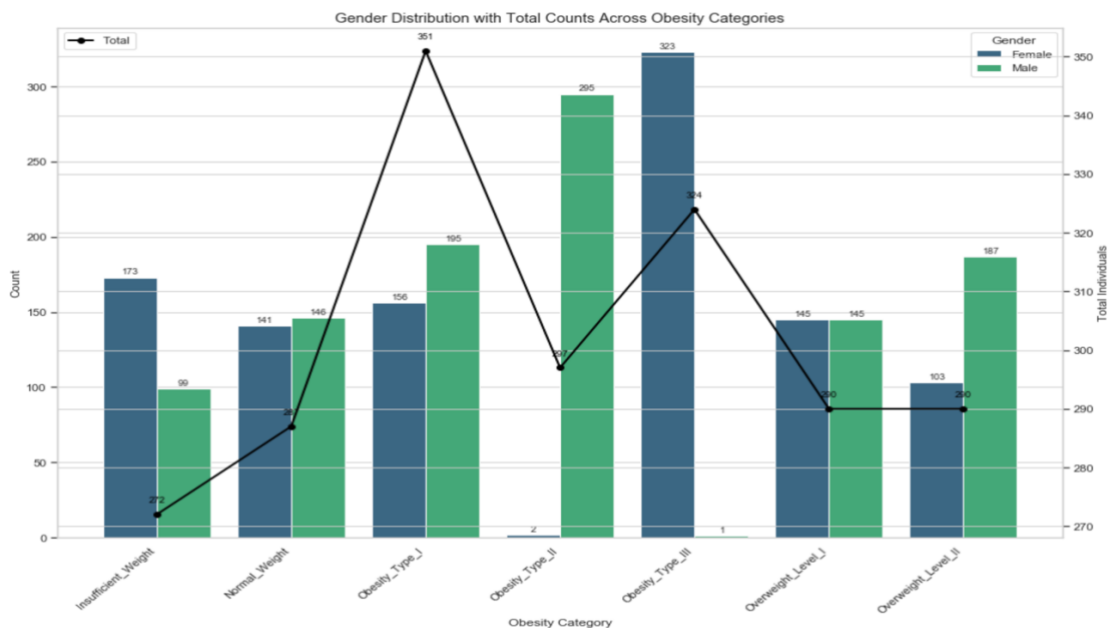
- Age is moderately negatively correlated with TUE (-0.30), indicating younger individuals spend more time on technology.
- Height is positively correlated with weight (0.46), FAF (0.29), and NCP (0.24).
- Weight shows only a weak correlation with FCVC (0.22).
- Many lifestyle variables show near-zero correlation with weight and age, indicating low multicollinearity and suitability for machine learning.



4.4 Numerical vs Categorical Analysis

Bar plots and box plots revealed:

- Mean weight increases steadily across obesity categories.
- Physical activity (FAF) decreases as obesity severity increases.
- Alcohol consumption and smoking show limited direct influence on obesity levels.
- Gender differences become pronounced at extreme obesity levels.



5. PRINCIPAL COMPONENT ANALYSIS (PCA)

PCA was applied to numerical features for visualization purposes.

5.1 Selection of Components

Two principal components were selected to enable 2D visualization:

- PC1: Dominated by weight, height, and age, representing body size.
- PC2: Influenced by lifestyle behaviors such as physical activity, meal frequency, hydration, and technology use.

5.2 PCA Interpretation

PC1 separates individuals primarily based on body composition, while PC2 captures lifestyle variation. Obesity categories are visibly separated along PC1, confirming the dominance of physical attributes, while PC2 adds behavioral nuance.

6. SUPERVISED MACHINE LEARNING MODELS

Five supervised models were implemented:

1. Logistic Regression
2. K-Nearest Neighbors (KNN)
3. Decision Tree
4. Random Forest
5. Support Vector Machine (SVM)

6.1 Evaluation Metrics Calculated

- Accuracy
- Precision
- Recall
- F1-score
- Macro F1-score (primary metric)

Macro F1-score was chosen because it treats all obesity categories equally, ensuring fair evaluation across classes of different sizes.

7. MODEL RESULTS & PERFORMANCE

Model	Accuracy	Macro F1-Score	Key Observations
Logistic Regression	0.8747	0.8708	Strong baseline model; performs well overall but struggles with boundary classes such as Normal Weight vs Overweight Level I.
K-Nearest Neighbors (KNN)	0.8251	0.8131	Lowest-performing model due to overlapping class regions and sensitivity to distance in high-dimensional space.

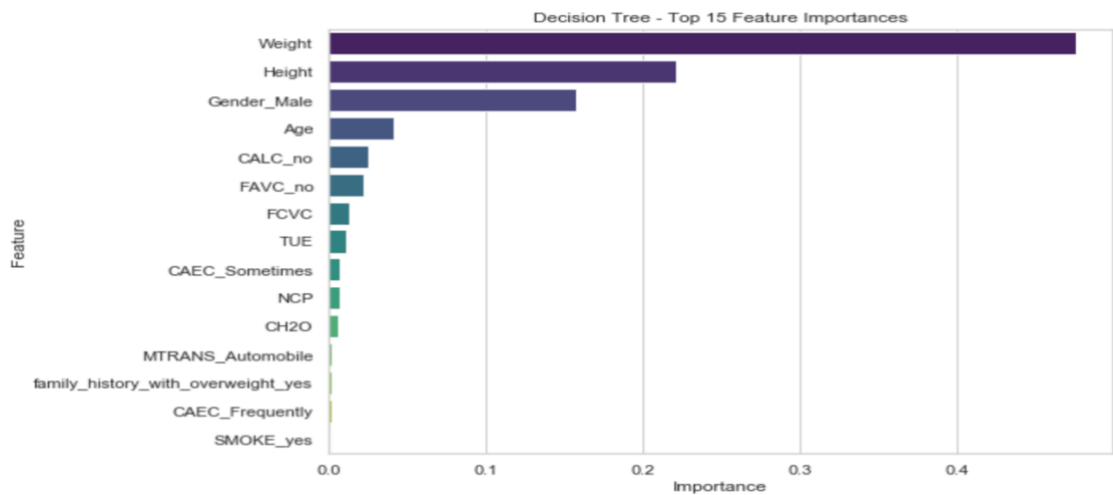
Model	Accuracy	Macro F1-Score	Key Observations
Decision Tree	0.9149	0.9133	Captures nonlinear relationships effectively; however, as a single tree, it shows mild overfitting tendencies.
Random Forest	0.9362	0.9361	Best-performing model; provides high accuracy, strong generalization, and balanced performance across all obesity categories.
Support Vector Machine (SVM)	0.9243	0.9230	Performs consistently well with strong class separation, especially effective in high-dimensional feature space.

8. FEATURE IMPORTANCE ANALYSIS

8.1 Decision Tree Feature Importance

Top features:

- Weight
- Height
- Gender
- Age



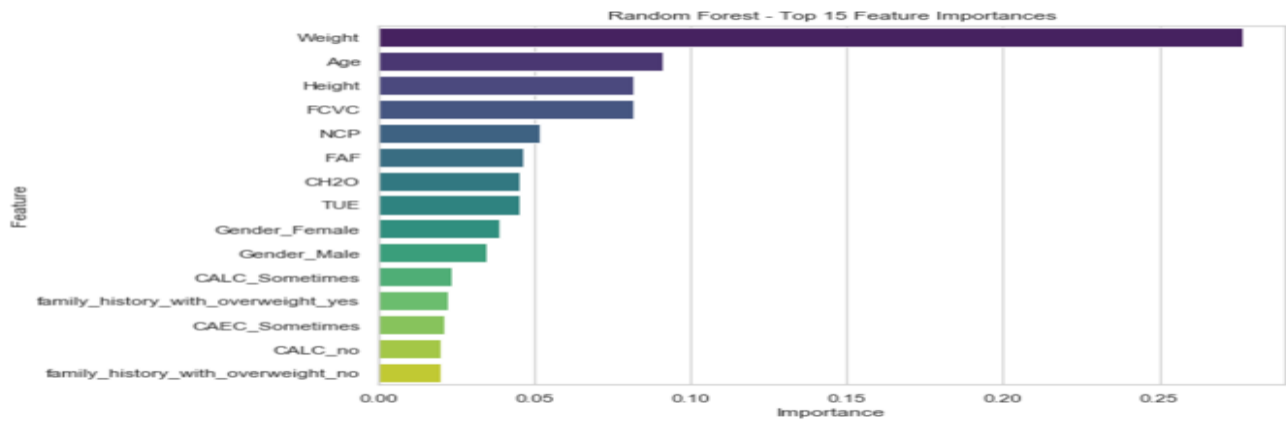
The Decision Tree relied heavily on a small number of dominant features.

8.2 Random Forest Feature Importance

Top features:

- Weight
- Age
- Height
- FCVC
- NCP
- FAF
- CH2O

Random Forest distributed importance more evenly, highlighting the contribution of lifestyle factors in addition to physical attributes.



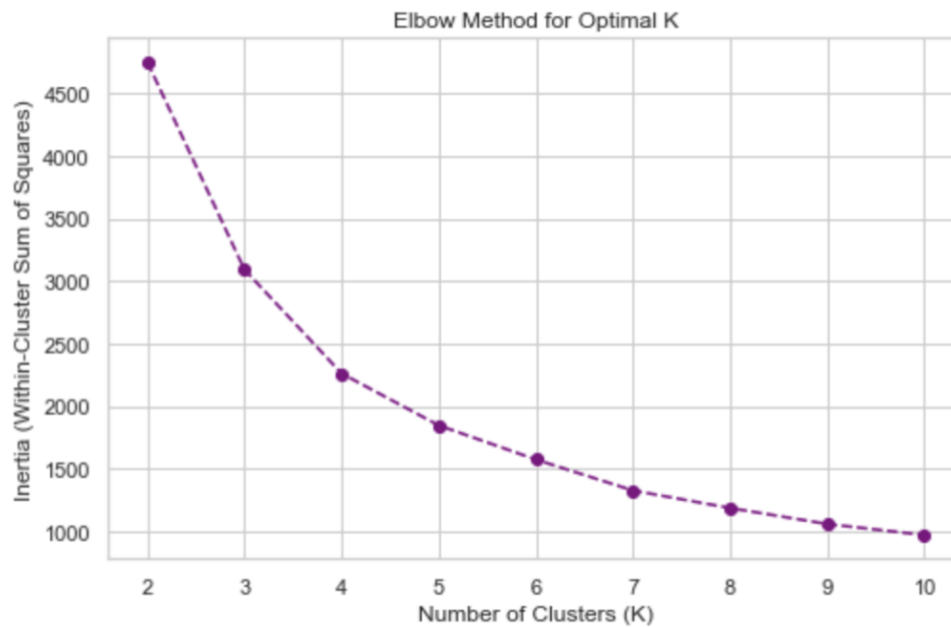
9. ROC CURVE ANALYSIS

ROC curves were generated using a one-vs-rest approach.

- All models produced curves close to the top-left corner.
- Random Forest achieved near-perfect AUC values for most classes.
- ROC curves confirmed strong separability between obesity categories.

10. UNSUPERVISED ML MODEL : K-MEANS CLUSTERING

The Elbow Method identified **K = 4** as the optimal number of clusters.



10.1 Cluster Profiling

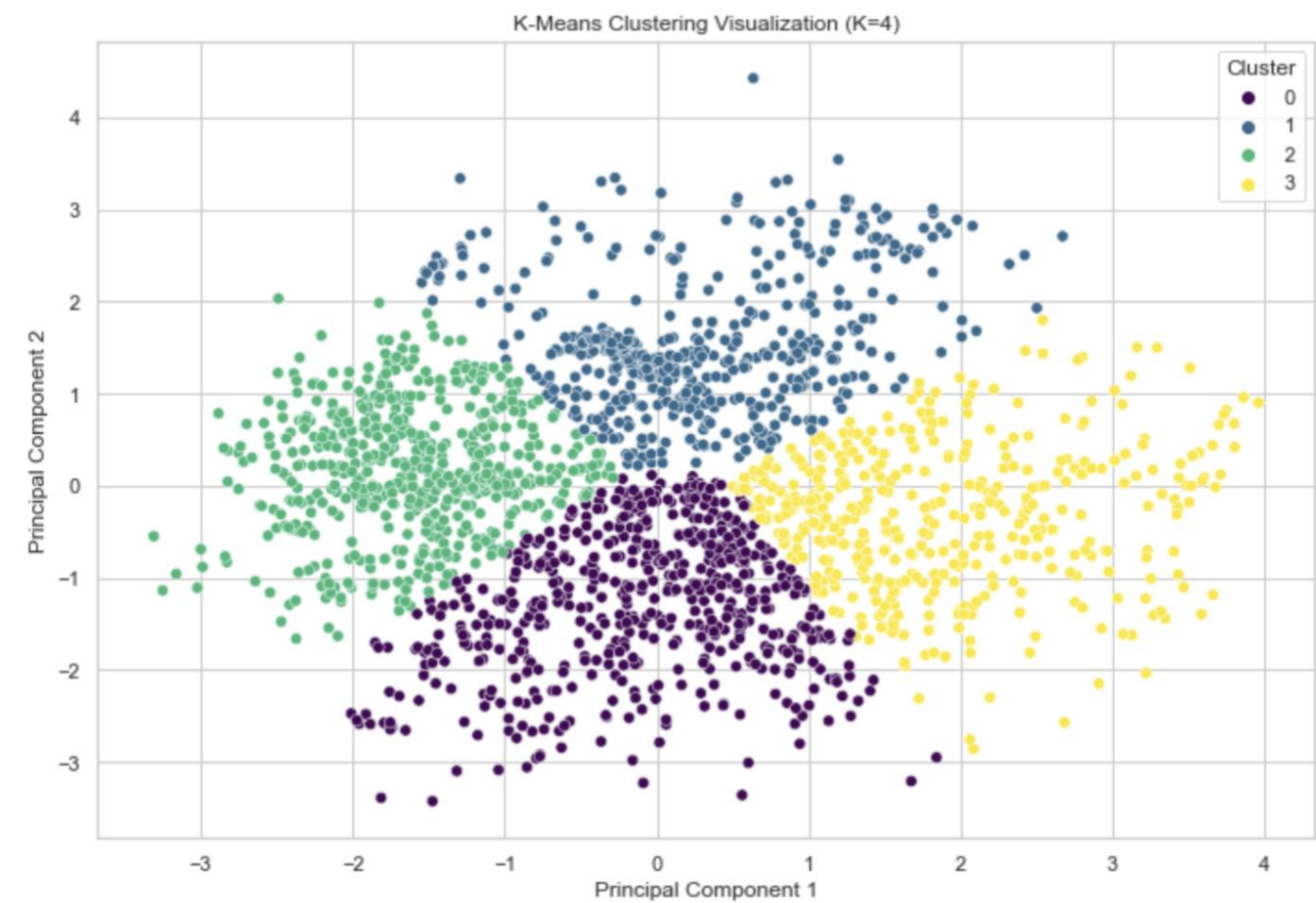
Cluster profiling was performed manually by analyzing cluster means and obesity label distributions.

- **Cluster 0:** Healthy, lean, active young adults
- **Cluster 1:** Moderately obese, sedentary adults
- **Cluster 2:** Severely obese individuals with poor activity patterns
- **Cluster 3:** Young, light-weight individuals

These clusters revealed natural lifestyle-based groupings independent of predefined labels.

col_0	0	1	2	3
NObeyesdad				
Insufficient_Weight	147	3	11	111
Normal_Weight	117	27	15	128
Obesity_Type_I	103	86	100	62
Obesity_Type_II	2	68	208	19
Obesity_Type_III	1	184	139	0
Overweight_Level_I	104	50	46	90
Overweight_Level_II	96	87	62	45

Clusters Formed based on K-Means Clustering:



11. KEY FACTORS CONTRIBUTING TO OBESITY

Based on EDA, supervised modeling, and clustering:

1. Weight (strongest predictor)
2. Height
3. Age
4. Physical activity (FAF)
5. Meal frequency (NCP)
6. Vegetable consumption (FCVC)
7. Hydration (CH2O)
8. Family history of overweight
9. Gender (moderate influence)

Smoking showed minimal influence in this dataset.

12. CONCLUSION

This project demonstrates that machine learning models can accurately predict obesity levels using physical and lifestyle-related data. Random Forest emerged as the most effective supervised model, while K-Means clustering provided valuable insights into natural lifestyle patterns. The results confirm that obesity is a multifactorial condition influenced by body composition, behavior, and demographics.

13. FUTURE WORK

Binary Classification (Obese vs Not Obese):

Future work can simplify the problem into a binary classification task to support faster clinical screening and real-world decision making.

Longitudinal Obesity Tracking:

Incorporating time-series or longitudinal data would enable tracking obesity progression and identifying early risk patterns over time.

Advanced Ensemble Methods (XGBoost):

More advanced boosting algorithms could be explored to further improve predictive accuracy and computational efficiency.

Personalized Intervention Modeling:

Machine learning can be extended to generate individualized lifestyle recommendations based on a person's dominant obesity risk factors.

14. REFERENCES:

<https://www.sciencedirect.com/science/article/pii/S2352340919306985?via%3Dihub>

<https://www.semanticscholar.org/paper/Dataset-for-estimation-of-obesity-levels-based-on-Palechor-Manotas/35b40bacd2ffa9370885b7a3004d88995fd1d011>